Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

APPLIED BUSINESS

Practical – VIII

Roll No.: T019	Name: Krishna Mishra
Class: TYIT	Batch: 1
Date of Assignment: 11-09-26	Date/Time of Submission: 11-09-26

Dataset: supermarket_sales.csv

Part A: Confidence Interval Estimation

1. Load the supermarket_sales.csv dataset into a Pandas DataFrame.
2. Display the first five records and check the size of the dataset.
3. Select the revenue column for analysis.
4. Check and handle missing values in the selected column.
5. Calculate the sample size, sample mean, and sample standard deviation.
6. Calculate the 90% confidence interval for the population mean revenue.
7. Calculate the 95% confidence interval for the population mean revenue.
8. Calculate the 99% confidence interval for the population mean revenue.
9. Display the confidence intervals in a suitable table.
10. Compare the 90%, 95%, and 99% confidence intervals.
11. Create a suitable visualization to display the sample mean and confidence interval.
12. Interpret the 95% confidence interval in the context of supermarket revenue.
13. Select the customer_type column and calculate the proportion of customers belonging to each category.
14. Calculate the 95% confidence interval for the proportion of Member customers.
15. Interpret the confidence interval for the population proportion in a business context.

CODE

```
import pandas as pd
```

```
import numpy as np

import matplotlib.pyplot as plt
from scipy import stats

from statsmodels.stats.proportion import proportion_confint



# Q1. Load dataset

df = pd.read_csv("supermarket_sales.csv")



# Q2. First 5 records and size

print("Q2:")

print(df.head())

print("Size:", df.shape)



# Q3 & Q4. Revenue and missing values

revenue = df["revenue"].dropna()

print("\nQ4. Missing values:", df["revenue"].isnull().sum())



# Q5. Sample statistics

n = len(revenue)

mean = revenue.mean()

std = revenue.std()



print("\nQ5:")

print("Sample Size:", n)

print("Mean:", round(mean, 2))
```

```
print("Standard Deviation:", round(std, 2))



# Q6-Q8. Confidence Intervals
cis = {}

for level in [90, 95, 99]:

 ci = stats.t.interval(

 level/100,

 df=n-1,

 loc=mean,

 scale=std/np.sqrt(n)

 )

 cis[level] = ci

 print(f"\nQ{level}% Confidence Interval:",

 round(ci[0], 2), "to", round(ci[1], 2))



# Q9. Confidence interval table

table = pd.DataFrame(cis, index=["Lower", "Upper"]).T

print("\nQ9. CI Table:")

print(table.round(2))



# Q10. Comparison

print("\nQ10: Higher confidence level gives a wider interval.")



# Q11. Visualization

plt.errorbar(
```

```
[90, 95, 99],

[mean, mean, mean],

yerr=[

[mean-cis[x][0] for x in [90,95,99]],
[cis[x][1]-mean for x in [90,95,99]]

],

fmt='o',

capsize=5

)

plt.axhline(mean, linestyle='--', label="Sample Mean")

plt.xlabel("Confidence Level")

plt.ylabel("Revenue")

plt.title("Revenue Confidence Intervals")

plt.legend()

plt.show()



# Q12. 95% Interpretation

print("\nQ12:")

print("95% CI means the true average revenue is expected")

print("to lie between", round(cis[95][0],2), "and", round(cis[95][1],2))



# Q13. Customer proportions

prop = df["customer_type"].value_counts(normalize=True)

print("\nQ13. Customer Proportions:")

print((prop*100).round(2), "%")
```

```
# Q14. 95% CI for Member proportion

members = (df["customer_type"] ==

"Member").sum() total =

df["customer_type"].notna().sum()

member_prop = members / total
ci_prop = proportion_confint(members, total, alpha=0.05)



print("\nQ14:")

print("Member Proportion:", round(member_prop*100, 2),

"%") print("95% CI:", round(ci_prop[0]*100, 2), "% to",

round(ci_prop[1]*100, 2), "%")



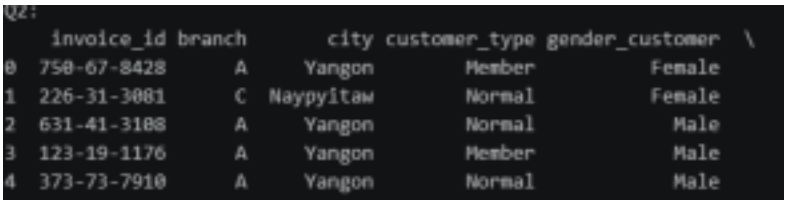
# Q15. Interpretation

print("\nQ15:")

print("We are 95% confident that the population proportion")

print("of Member customers lies within the above interval.")
```

OUTPUT



Q2:

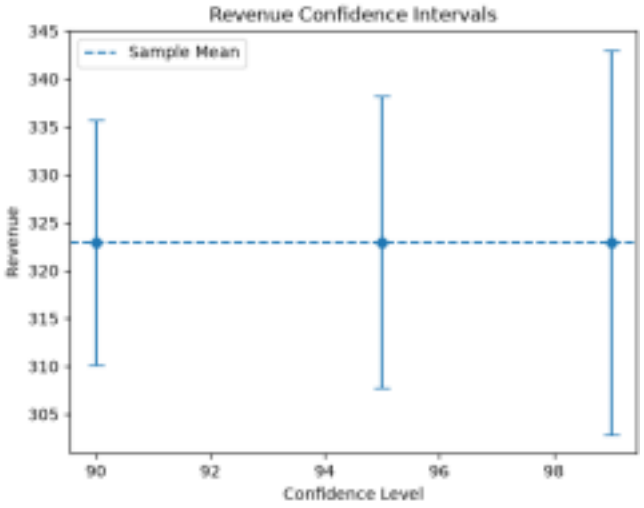
	invoice_id	branch	city	customer_type	gender_customer	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	

	product_line	unit_cost	quantity	%pct_markup	revenue	\
0	Health and beauty	74.69	7	26.1415	548.9715	
1	Electronic accessories	15.28	5	3.8200	80.2200	
2	Home and lifestyle	46.33	7	16.2155	340.5255	
3	Health and beauty	58.22	8	23.2880	489.0480	
4	Sports and travel	86.31	7	30.2085	634.3785	

```
        date   time payment_method    cogs    gm_pct  gross_income  rating
0   01/05/19  13:08        Ewallet  522.83  4.761905       26.1415     9.1
1   03/08/19  10:29           Cash   76.40  4.761905        3.8200     9.6
2   03/03/19  13:23    Credit card  324.31  4.761905       16.2155     7.4
3  1/27/2019  20:33        Ewallet  465.76  4.761905       23.2880     8.4
4   02/08/19  10:37        Ewallet  604.17  4.761905       30.2085     5.3
Size: (1000, 17)
```

```
Q4. Missing values: 0
...
95  307.71  338.23
99  302.90  343.03

Q10: Higher confidence level gives a wider interval.
```




```
Q12:
95% CI means the true average revenue is expected
to lie between 307.71 and 338.23

Q13. Customer Proportions:
customer_type
Member    50.1
Normal    49.9
Name: proportion, dtype: float64 %

Q14:
Member Proportion: 50.1 %
95% CI: 47.0 % to 53.2 %

Q15:
We are 95% confident that the population proportion
of Member customers lies within the above interval.
```

Part B: Bootstrapping

16. Select the revenue column and calculate its original sample median.
17. Create a bootstrap sample by randomly selecting observations from the original sample with replacement.
18. Calculate the median of the bootstrap sample.
19. Repeat the bootstrapping process 1,000 times and store the median from each bootstrap sample.
20. Create a histogram of the bootstrap median values.
21. Calculate the 2.5th percentile and 97.5th percentile of the bootstrap median values.

22. Use these percentile values to calculate the 95% bootstrap confidence interval for the population median.
23. Display the original sample median and the bootstrap confidence interval.
24. Compare the original median with the bootstrap confidence interval.
25. Interpret the bootstrap confidence interval in the context of supermarket revenue.
26. Write a conclusion explaining how confidence intervals and bootstrapping can help businesses make decisions using sample data.

CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("supermarket_sales.csv")

# Q16. Original sample median
revenue = df["revenue"].dropna()
original_median = revenue.median()

print("Q16. Original Sample Median:", round(original_median, 2))

# Q17-Q19. Bootstrap 1000 samples and calculate medians
boot_medians = []

for i in range(1000):
 sample = np.random.choice(revenue, size=len(revenue), replace=True)
boot_medians.append(np.median(sample))

print("\nQ17-Q19. Bootstrap completed.")
print("Number of Bootstrap Samples:", len(boot_medians))

# Q20. Histogram
plt.hist(boot_medians, bins=30, edgecolor="black")
plt.axvline(original_median, linestyle="--", label="Original Median")
plt.xlabel("Bootstrap Median")
plt.ylabel("Frequency")
plt.title("Bootstrap Distribution of Revenue Median")
plt.legend()
plt.show()

# Q21-Q22. 95% Bootstrap Confidence Interval
lower = np.percentile(boot_medians, 2.5)
upper = np.percentile(boot_medians, 97.5)

print("\nQ21-Q22. 95% Bootstrap Confidence Interval:")
print("Lower:", round(lower, 2))
print("Upper:", round(upper, 2))

# Q23. Display median and CI
```

```
print("\nQ23:")
print("Original Median:", round(original_median, 2))
print("95% Bootstrap CI:", round(lower, 2), "to", round(upper, 2))

# Q24. Compare
print("\nQ24:")
print("The original median lies within the bootstrap confidence interval.")

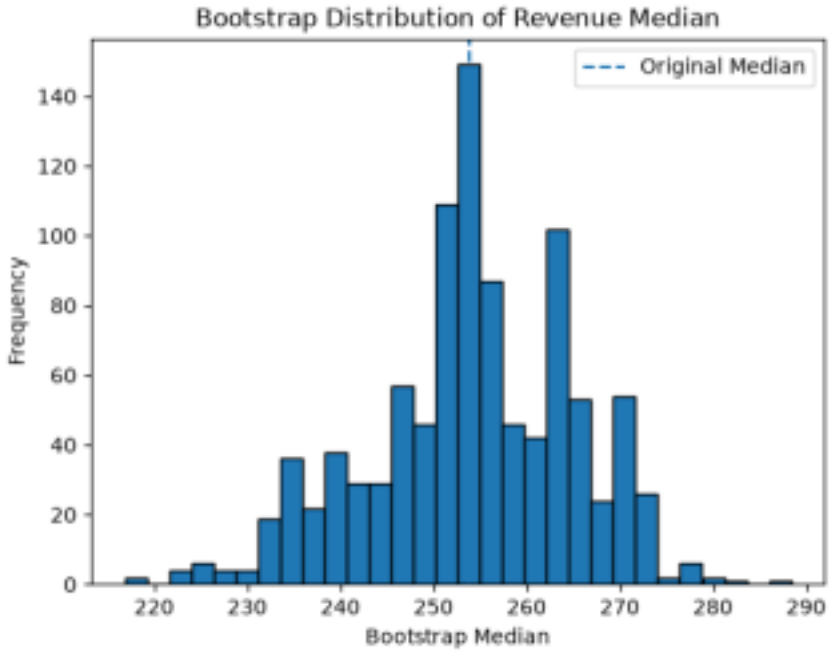
# Q25. Interpretation
print("\nQ25:")
print("We are 95% confident that the population median")
print("revenue lies between", round(lower, 2), "and", round(upper, 2))

# Q26. Conclusion
print("\nQ26. Conclusion:")
print("Confidence intervals and bootstrapping help businesses")
print("estimate population values using sample data.") print("They
help businesses make better decisions under uncertainty.")
```

OUTPUT

```
Q16. Original Sample Median: 253.85

Q17-Q19. Bootstrap completed.
Number of Bootstrap Samples: 1000
```



```
Q21-Q22. 95% Bootstrap Confidence Interval:
Lower: 232.64
Upper: 272.69

Q23:
Original Median: 253.85
95% Bootstrap CI: 232.64 to 272.69

Q24:
The original median lies within the bootstrap confidence interval.

Q25:
We are 95% confident that the population median
revenue lies between 232.64 and 272.69

Q26. Conclusion:
Confidence intervals and bootstrapping help businesses
estimate population values using sample data.
They help businesses make better decisions under uncertainty.
```